

Structural Organisation in Animals

1. In male cockroach, genital pouch is present at the hind end of abdomen which is bounded

- (1) Dorsally by 9th and 10th terga and ventrally by the 9th sternum
- (2) Dorsally by 9th tergum and ventrally by 9th and 10th sternum
- (3) Dorsally by 8th tergum and ventrally by 9th sternum
- (4) Dorsally by 10th tergum and ventrally by 10th sternum

2. Size ranges of cockroach from

- (1) 0.25-3 inches
- (2) 0.6-7.6 cm
- (3) 34-53 mm
- (4) All

3. Read the following statements

- (A) Cockroach consists of a series of fused, segmentally arranged ganglia**
- (B) Ganglia joined in cockroach by paired longitudinal connectives on the ventral side**
- (C) Nervous system present on dorsal side of cockroach**
- (D) In cockroach, total no. of ganglia is Nine in number.**

- (1) All statements are correct
- (2) Only A, B and C are correct
- (3) Only A, B and D are correct
- (4) Only B, C and D are correct

4. Which structure help in grinding of food

- (1) Gizzard
- (2) Proventriculus
- (3) Mandible
- (4) All

5. Match the column-I with column-II and select the correct option (in cockroach):

Column-I	Column-II
(A) Mushroom gland	(P) 6th segment
(B) Testis	(Q) 2nd – 6th segments
(C) Spermatheca	(R) 4th – 6th segments
(D) Ovaries	(S) 6th –7th segments

- (1) (A)–S, (B)–R, (C)–P, (D)–Q
- (2) (A)–S, (B)–R, (C)–Q, (D)–P
- (3) (A)–R, (B)–S, (C)–P, (D)–Q
- (4) (A)–S, (B)–P, (C)–R, (D)–Q

6. Which statements is correct about cockroach

- (1) Few species thrive in and around human habitat
- (2) They are pests because they destroy food and contaminate it with their smelly excreta
- (3) Many species of cockroaches are wild and are many economic importance
- (4) All statements are correct

7. How many of the following structures is/are associated with excretion in male cockroach?

Choose the correct option.

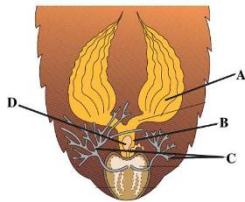
Ureose glands, Malpighian tubules, Fat body, Collateral glands, Nephrocytes

- (1) Five
- (2) One
- (3) Four
- (4) Three

8. In female cockroach two large ovaries, lying

- (1) Ventrally 2nd – 6th segments
- (2) Dorsally 2nd – 6th segments
- (3) Laterally 4th – 6th segments
- (4) Laterally 2nd – 6th segments

9. Which structure help in the formation of ootheca.



- (1) A (2) B
(3) C (4) D

10. Select the incorrect statement from the following w.r.t. *Periplaneta americana*.

- (1) The hindgut is broader than the midgut and is differentiated into ileum, colon and rectum.
(2) Gizzard helps in grinding the food particles.
(3) A ring of 100 – 150 blind tubules called gastric caeca is present at the junction of foregut and midgut.
(4) The oesophagus opens into a sac-like structure called crop for storing of food.

11. On an average, a female cockroach produces ____ oothecae, each containing ____ eggs.
Choose the option which correctly fills the respective blanks.

- (1) 4-6, 7-8
(2) 14-16, 9-10
(3) 9-10, 14-16
(4) 7-8, 4-6

12. Assertion : Visceral organs located in haemocoel are bathed in haemolymph in *Periplaneta*.
Reason : Cockroach has an open circulatory system and blood in haemocoel transports the oxygen to different parts of body.

Choose the correct answer.

- (1) Both Assertion and Reason are true and reason is the correct explanation of the assertion
(2) Both Assertion and Reason are true but reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

13. The respiratory system of cockroach consists of a network of A , that open through B of small holes, present on the C side of the body.
Select the option which fills these blanks correctly.

- | A | B | C |
|-------------|----------|---------|
| (1) Trachea | 10 pairs | Ventral |
| (2) Gills | 8 pairs | Lateral |
| (3) Trachea | 8 pairs | Dorsal |
| (4) Trachea | 10 pairs | Lateral |

14. Read the following statements

Assertion : Cockroach is uricotelic

Reason : Malpighian tubules absorb nitrogenous waste products and convert them into uric acid which is excreted out through the hindgut.

Choose the correct option.

- (1) Assertion is correct, reason is correct; reason is a correct explanation for assertion.
(2) Assertion is correct, reason is correct; reason is not a correct explanation for assertion.
(3) Assertion is correct, reason is incorrect
(4) Assertion is incorrect, reason is correct

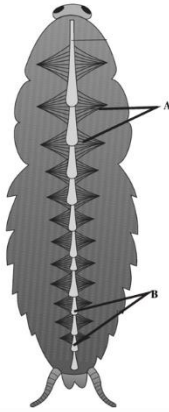
15. Which statement is incorrect about ootheca

- (1) Dark reddish to blackish -Brown colour
(2) Size is about 3/8" long
(3) Males produce 9–10 Ootheca
(4) Each ootheca containing 14–16 eggs (chamber)

16. Select the correct statement given below- (about cockroach)

- (1) A pair of thread like antennae arise from membranous sockets lying in front of Anus
(2) Antennae have sensory receptors that help in monitoring the environment.
(3) Anterior end of the head bears appendages forming the sucking and chewing type of mouth parts.
(4) All of these

17. Which of the following option is correct regarding A and B



- (1) A = Heart chamber B = Allary muscle
- (2) A = 13 in number B = 12 pairs
- (3) A = Allary muscle B = 13 in number
- (4) A = Heart chamber B = 12 pairs in number

18. Sclerite joined to each other by (in cockroach)

- (1) Thin and flexible articular membrane
- (2) Thin and non-flexible Arthroidal membrane
- (3) Thick and flexible Arthroidal membrane
- (4) Thick and non-flexible Articular membrane

19. True for reproduction in *Periplaneta americana*

- (i) Anterior part of female genital pouch contains female gonopore, spermathecal pores and collateral gland.
- (ii) In males, genital pouch or chamber lies at the hind end of abdomen bounded dorsally by 9th and 10th terga and ventrally by the 9th sternum
- (iii) Males genital pouch contains dorsal anus, ventral male genital pore and gonapophysis.
- (iv) Males bear a pair of short, thread like anal styles which are absent in females.
- (v) In both sexes, the 10th segment bears a pair of jointed filamentous structures called anal cerci.

Choose the correct option.

- (1) Only (i), (iii)
- (2) Only (i), (ii), (iii)
- (3) Only (i) (iii), (iv)
- (4) All

20. In Cockroach, outer layer of the Gizzard is made up of

- (1) Thick circular chitinous
- (2) Thick circular muscle
- (3) Thin circular chitinous
- (4) Thin circular muscle

21. Which of the following statements is not true w.r.t frogs?

- (1) Their body colour offers them protective colouration
- (2) Summer sleep of frog is called aestivation
- (3) Tail is present in their larval stage
- (4) Their mouth possess a pair of tongue

22. Match the column-I and column-II and select the correct option.

Column-I	Column-II
(a) Sensory Papillae	(p) Hearing
(b) Nasal epithelium	(q) Touch
(c) Buds	(r) Smell
(d) Tympanum	(s) Taste

Choose the correct option.

- (1) (a)–q, (b)–r, (c)–s, (d)–p
- (2) (a)–q, (b)–r, (c)–p, (d)–s
- (3) (a)–r, (b)–q, (c)–s, (d)–p
- (4) (a)–q, (b)–s, (c)–q, (d)–p

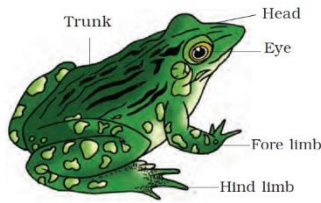
23. In frog, the colour of dorsal side of body is generally

- (1) Olive green
- (2) Pale yellow
- (3) Dark yellow
- (4) Plant green

24. In male frogs, a small, median chamber that is used to pass faecal matter, urine and sperms to the exterior is

- (1) Fat body
- (2) Cloaca
- (3) Urinary bladder
- (4) Ureter

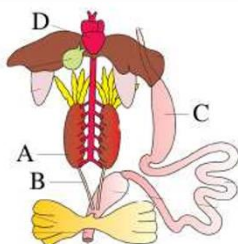
25. Which of the following part in the given diagram have copulatory pad



- (1) Head
(2) Trunk
(3) Fore limb
(4) Hind limb
26. Male frogs can be distinguished from female frogs on the basis of presence of
- Webbed digits in hind limbs
 - Sound producing vocal sacs
 - A copulatory pad on first digit of the forelimbs.
 - A copulatory pad on last digit of the forelimbs.

Choose the correct option.

- (1) a and b
(2) b and c
(3) c and d
(4) a and d
27. At a time mature female frog can lay.....ova at a time
- (1) 2000-2500
(2) 2500-3000
(3) 1000-2000
(4) 1500-2000
28. Which of the following part is common part for urine and gametes transport.



- (1) A
(2) B
(3) C
(4) D

29. In frog function of liver is

- (1) emulsification of fat
(2) provide alkaline medium to food
(3) digestion of protein
(4) both (1) and (2)

30. The lymphatic system of frogs consists of

- (a) Lymph
(b) Lymph channels
(c) Lymph nodes

Select the option with correct set.

- (1) (a) and (b) only
(2) (a) and (c) only
(3) (a) only
(4) (a), (b) and (c)

31. In frog, the medulla oblongata passes out through the.....(a).... and continues into(b)... a and b respectively

- (1) Spinal cord, foramen magnum
(2) Spinal cord, vertebral column
(3) Foramen magnum, Spinal cord
(4) Vertebral column, spinal cord

32. Male reproductive system of frog consist of

- (1) A pair of red coloured testes
(2) A pair of yellowish ovoid testes
(3) A pair of rounded testes
(4) Two pair of rounded testes

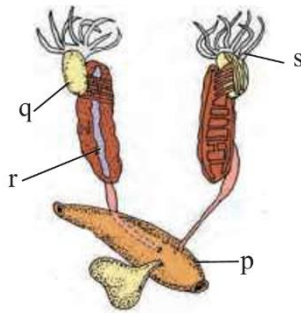
33. Testes attached with kidney with the help of

- (1) Double fold of peritoneum
(2) Mesorchium
(3) Reticular fibres
(4) (1) and (2) both

34. Select the correct statement from below regarding frog :

- (1) They show aestivation and hibernation
(2) skin is smooth and slippery
(3) They do not have the constant body temperature
(4) all of these

35. Read the following diagram and Match the column-I with column- II



Column-I	Column-II
(A) Testis	(i) p
(B) Adrenal gland	(ii) q
(C) Cloaca	(iii) r
(D) Fat Bodies	(iv) s

Choose the correct option.

- (1) A-ii; B-iii; C-i; D-iv
- (2) A-ii; B-iii; C-iv; D-i
- (3) A-ii; B-i; C-iii; D-iv
- (4) A-i; B-ii; C-iii; D-iv

36. Read the following carefully

- (a) Secrete elastin fibres
- (b) Secrete modified polysaccharides that act as ground substance
- (c) Present in areolar tissue
- (d) Absent in the main fluid connective tissue

All of the above are true for

- (1) Mast cells
- (2) Macrophages
- (3) Fibroblasts
- (4) Leucocytes

37. All of the following statements are correct w.r.t. bones, except

- (1) Serve weight-bearing functions
- (2) Ground substance is rich in calcium salts and collagen fibres
- (3) Concentric lamellae are present in their matrix
- (4) Have osteocytes that are always present outside the lacunae

38. If a single neuron is suitably stimulated by a threshold stimulus, an electrical disturbance will travel

- (1) Away from the cell body towards dendrites through neurotransmitters
- (2) Irrespective of any direction
- (3) Towards axon terminals along its plasma membrane
- (4) Towards its cell body through axoplasm only.

39. Consider a hypothetical scenario where site of blood cell formation in an adult man is damaged. Which of the following can be a probable option of treatment in the above described case?

- (1) Administration of only immature osteocytes
- (2) Blood transfusion only for once
- (3) Bone marrow transplantation
- (4) Periodic infusion of adipocytes

40. Read the following statements carefully.

Statement (A): All involuntary muscle fibres are unstriated.

Statement (B): All the striated muscle fibres are voluntary.

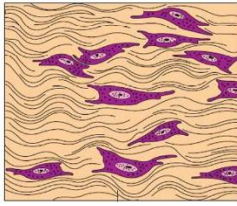
Select the correct option.

- (1) Both statements (A) and (B) are correct
- (2) Both statements (A) and (B) are incorrect
- (3) Only statement (A) is correct
- (4) Only statement (B) is correct

41. The epithelial tissue which is made up of many layers of cells

- (1) Plays a main role in the secretion and absorption
- (2) Has a limited role in providing protection against the mechanical stress
- (3) Covers the moist surface of buccal cavity and pharynx
- (4) Is involved in the formation of a diffusion boundary in blood vessels

42. Which of the following option is incorrect regarding the diagram



- (1) fibres show a regular pattern
- (2) collagen fibres are present in rows
- (3) fibres and fibroblasts are compactly packed
- (4) All statement are correct

43. Which of the following is true for neural tissue present in humans?

- (1) Has excitable cells called neuroglia
- (2) More than 50% of this tissue is constituted by neurons
- (3) Neuroglial cells are the structural and functional units of this tissue.
- (4) Neurons do not have the ability to produce stimulus.

44. Consider the following carefully.

- (A) Unstriated and involuntary muscle fibres present in the walls of internal organs.
- (B) Contractile tissue present only in the human heart
- (C) Cells in the tissue having a free surface and held together with little intercellular material.

Which of the above tissues show the presence of cell junctions?

- (1) (A) and (C) only
- (2) (B) and (C) only
- (3) (A), (B) and (C)
- (4) (A) and (B) only

45. Which tissue found in the inner surface of hollow organs like bronchioles and fallopian tube.

- (1) Ciliated columnar epithelium
- (2) Ciliated cuboidal epithelium
- (3) Compound epithelium
- (4) All of these

46. Which tissue found in the lining of stomach and intestine.

- (1) Simple squamous epithelium
- (2) Simple cuboidal epithelium
- (3) Simple columnar epithelium
- (4) Compound columnar epithelium

47. Assertion : Adipose tissue helps to store fat

Reason : Tendon is the example of dense connective tissue

- (1) Assertion is correct, reason is correct; reason is a correct explanation for assertion.
- (2) Assertion is correct, reason is correct; reason is not a correct explanation for assertion.
- (3) Assertion is correct, reason is incorrect
- (4) Assertion and reason both are incorrect

48. Most abundant and widely distributed tissue is

- (1) Epithelial
- (2) Connective
- (3) Muscular
- (4) Nervous

49. Cell of wall of blood vessels is composed of

- (1) Simple cuboidal connective tissue
- (2) Simple squamous connective tissue
- (3) Simple squamous epithelium
- (4) Stratified cuboidal epithelium

50. A teacher in the class once asked about the location of smooth muscles in the human body, to which the students gave various examples that are listed below:

- | | |
|----------------------------|------------------------|
| (a) Walls of blood vessels | (b) Walls of heart |
| (c) Stomach wall | (d) Walls of intestine |
| (e) Biceps | |

Choose the correct set of examples given by the students. w.r.t. location of smooth muscles.

- (1) (a), (c) and (d)
- (2) (b), (c) and (d)
- (3) (a), (b) and (c)
- (4) (c), (d) and (e)

Answer Key

1. (1)
2. (4)
3. (3)
4. (4)
5. (1)
6. (2)
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43. (4)
44. (3)



- 45. (1)
- 46. (3)
- 47. (2)
- 48. (2)
- 49. (3)
- 50. (1)

